

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250271151

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

DeRossett; Andrew Q.

PACKAGED TERMINAL AIR CONDITIONER WITH RECEPTACLE WIRELESS POWER SYSTEM

Abstract

A packaged terminal air conditioner with a receptacle wireless power system. The PTAC wireless power system mountable to a wall and includes a wireless power transmitter capable of powering a wirelessly powered PTAC positioned proximate thereto.

Inventors: DeRossett; Andrew Q. (Shelbyville, KY)

Applicant: Midea Group Co., Ltd. (Foshan, CN)

Family ID: 1000007725902

Assignee: Midea Group Co., Ltd. (Foshan, CN)

Appl. No.: 18/586176

Filed: February 23, 2024

Publication Classification

Int. Cl.: F24F1/031 (20190101); F24F7/013 (20060101); F24F11/52 (20180101); F24F13/20 (20060101)

U.S. Cl.:

CPC F24F1/031 (20190201); F24F7/013 (20130101); F24F11/52 (20180101); F24F13/20 (20130101); F24F2013/207 (20130101)

Background/Summary

BACKGROUND

[0001] The present embodiments relate to a packaged terminal air conditioner and/or receptacle, more specifically a wireless power system for a packaged terminal air conditioner and/or receptacle.

[0002] A wide variety of packaged terminal air conditioner(s), hereinafter “PTACs”, are commonly used in residential and/or commercial applications. These PTAC heating and/or cooling units are often used in, but not limited to, hotels, motels, and other lodging establishments in the hospitality industry to provide comfortable and consistent temperature control for guests. Many of these PTACs are installed in an opening defined by a sleeve or receptacle that is permanently mounted in an outer wall of a building. A PTAC is often plugged in to a dedicated electrical outlet that is generally hidden from view when the PTAC is in its installed location with the receptacle. The power cords used by such PTACs, however, can be cumbersome and unsightly, both during use and in storage. In addition, PTACs may have different electrical requirements (e.g., 15, 20, or 30 amps and 208, 230, 265, or 277 voltages in the U.S., as well as different types of plugs that are only compatible with certain electrical outlets).

[0003] PTACs used in hospitality industry applications may need to be removed, serviced and/or replaced from time to time. However, given that individual PTAC units may have different electrical requirements, and that lodging establishments may install various types of PTACs, receptacles and electrical outlets in different rooms over time, it may be cumbersome for service personnel to swap out PTAC units in a lodging establishment, short of purchasing only PTACs with identical electrical requirements. Thus, there is a need to simplify the management of multiple PTACs within a lodging establishment.

SUMMARY

[0004] The herein-described embodiments address these and other problems associated with the art by providing a PTAC receptacle wireless power system, and that includes a wireless power transmitter capable of powering a wirelessly powered PTAC positioned proximate thereto. In some instances, the wireless power transmitter/receiver may be stationary, however the wireless power transmitter/receiver may be movable between storage (e.g. preinstallation) and use positions or configurations relative to a fixed remaining portion of the corresponding PTAC/receptacle.

[0005] In some embodiments, a PTAC wireless power system may include a PTAC receptacle. In various embodiments, the PTAC receptacle may be configured to mount to an outer wall. In some embodiments, the PTAC wireless power system may include a wireless power transmitter supported by the PTAC receptacle. In various embodiments, the wireless power transmitter may be configured to generate a wireless power signal to power a wirelessly powered PTAC positioned proximate thereto.

[0006] In addition, in some embodiments, the wirelessly powered PTAC may be movable between a deployed position and a stowed position relative to the PTAC receptacle. In various embodiments, the wireless power transmitter may be upwardly-facing to oppose a downwardly-facing wireless power receiver of the wirelessly powered PTAC when the wirelessly powered PTAC is in the stowed position. In some embodiments, the wireless power transmitter may be downwardly-facing to oppose an upwardly-facing wireless power receiver of the wirelessly powered PTAC when the wirelessly powered PTAC is in the stowed position. In various embodiments, the wireless power transmitter may be side-facing to oppose a side-facing wireless power receiver of the wirelessly powered PTAC when the wirelessly powered PTAC is in the stowed position. In some embodiments, the PTAC wireless power system may include a power cord configured to supply power to the wireless power transmitter from a wall-mounted electrical outlet. In various embodiments, the PTAC receptacle may be configured to supply power to the wireless power transmitter from a hard-wired connection of an electrical circuit. In some embodiments, the PTAC wireless power system may include a user interface and a controller

configured to operate the user interface and the wireless power transmitter. In various embodiments, a controller may be configured to control a functional operation of the wirelessly powered PTAC using a user interface. In some embodiments, the wirelessly powered PTAC may include a wireless power receiver, a user interface, and a controller configured to operate the user interface and the wireless power receiver. In various embodiments, a controller may be configured to control a functional operation of the PTAC receptacle using a user interface. In some embodiments, the wireless power transmitter may be configured to supply power to a wirelessly powered PTAC including an electric motor and/or a compressor. In various embodiments, the wirelessly powered PTAC may be lacking a power storage element sufficient to otherwise power the wirelessly powered PTAC. In some embodiments, the PTAC receptacle may include one or more members defining at least a portion of a through opening. In various embodiments, the wireless power transmitter may be supported by one or more members of the PTAC receptacle. In some embodiments, the one or more members may include a top wall, a bottom wall, and opposing side walls interconnecting the top wall and the bottom wall. In various embodiments, the PTAC wireless power system may include a controller configured to regulate a power output of the wireless power transmitter. In some embodiments, the controller may be configured to determine the power output required by the wirelessly powered PTAC and control the wireless power transmitter to output a wireless power signal that meets the power output required by the wirelessly powered PTAC. In various embodiments, the wireless power transmitter may be inwardly-facing to oppose an outwardly-facing wireless power receiver of the wirelessly powered PTAC.

[0007] In some embodiments, a PTAC wireless power system may include a wirelessly powered PTAC having a wireless power receiver. In various embodiments, the PTAC wireless power system may include a PTAC receptacle having a wireless power transmitter configured to generate a wireless power signal in communication with the wireless power receiver of the wirelessly powered PTAC to power the wirelessly powered PTAC positioned proximate thereto.

[0008] In addition, in some embodiments, the PTAC wireless power system may include a user interface and a controller configured to operate the user interface and the wireless power transmitter. In various embodiments, a controller may be configured to control a functional operation of the wirelessly powered PTAC using a user interface. In some embodiments, the wirelessly powered PTAC may include a wireless power receiver, a user interface, and a controller configured to operate the user interface and the wireless power receiver. In various embodiments, a controller may be configured to control a functional operation of the PTAC receptacle using a user interface. In some embodiments, the PTAC wireless power system may include a controller configured to regulate a power output of the wireless power transmitter. In various embodiments, a controller may be configured to determine the power output required by the wirelessly powered PTAC and control the wireless power transmitter to output a wireless power signal that meets the power output required by the wirelessly powered PTAC.

[0009] In some embodiments, a wirelessly powered PTAC may include a PTAC unit configured to couple with a PTAC receptacle between a stowed position and a deployed position. In various embodiments, the wirelessly powered PTAC may include a wireless power receiver supported by the wirelessly powered PTAC and configured to receive a wireless power signal provided by a wireless power transmitter of the PTAC receptacle to power the wirelessly powered PTAC when positioned in the stowed position.

[0010] Other embodiments may include various methods for making and/or using any of the aforementioned constructions.

[0011] These and other advantages and features, which characterize the invention, are set forth in the claims annexed hereto and forming a further part hereof. However, for a better understanding of the invention, and of the advantages and objectives attained through its use, reference should be made to the Drawings, and to the accompanying descriptive matter, in which there is described example embodiments of the invention. This summary is merely provided to introduce a selection

of concepts that are further described below in the detailed description, and is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used as an aid in limiting the scope of the claimed subject matter.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1 is a perspective view of a room incorporating a PTAC wireless power system consistent with some embodiments of the invention, illustrating a wirelessly powered PTAC in a stowed position relative to a PTAC receptacle used for providing wireless power.

[0013] FIG. 2 is a block diagram of an example control system for the PTAC wireless power system of FIG. 1.

[0014] FIG. 3 is a perspective view of a room incorporating a PTAC wireless power system, illustrating the wirelessly powered PTAC in a deployed position relative to the PTAC receptacle.

[0015] FIG. 4 is a side cross sectional view of the PTAC wireless power system of FIG. 1.

[0016] FIG. 5 is a perspective view of another embodiment of PTAC wireless power system, and used for providing wireless power through the side of a wirelessly powered PTAC and using a hard-wired electrical supply.

DETAILED DESCRIPTION

[0017] Turning now to the drawings, wherein like numbers denote like parts throughout the several views, FIG. 1 illustrates an example room (e.g., a hotel) **10** within which is installed a PTAC wireless power system **12** suitable for powering one or more wirelessly powered appliances, e.g., a wirelessly powered PTAC **60**. Room **10** includes one or more walls or structural elements (e.g. horizontal, vertical, ceiling, exterior, interior) **11**, which may be an outer or periphery wall of a building in some embodiments. The wall **11** may include window(s) **13** as shown in the one embodiment in communication with the exterior of the building or outdoors. However, other walls or building structure are contemplated. For example, PTACs and/or PTAC receptacles may or may not be secured to walls **11**, but may otherwise be generally floor standing such that their load is predominantly borne by the floor rather than a wall.

[0018] It will also be appreciated that room **10** may be any suitable living or working space within which may be provided including one or more walls or other structure, and within which it may be desired to use a wirelessly powered PTAC **60**. Example types of suitable rooms include a kitchen, bar, entertainment area, office area, retail establishment, commercial establishment, residential establishment, hospital, classroom, apartments, etc. In some embodiments, room **10** may even be disposed within a boat or recreational vehicle.

[0019] Room **10** may also include one or more electrical outlets **38**, e.g., disposed on walls or otherwise wall-mounted. In many rooms, for example, electrical outlets are positioned at various points along a wall **11** to provide users with multiple locations within which to plug in various small, portable and/or electronic devices.

[0020] PTAC wireless power system **12**, in the illustrated embodiments, is mounted to and supported at least in part by wall **11** although other applications are contemplated. In the embodiment of FIG. 1, for example, PTAC wireless power system **12** includes a PTAC/wall sleeve or receptacle **40** having a wireless power transmitter **44** (occluded by PTAC **60** in FIG. 1). A power cord **48**, plugged into an electrical outlet **38**, supplies power to PTAC wireless power system **12** or more specifically the receptacle **40**, although in other embodiments the PTAC wireless power system may be hard-wired. For example as shown in the one embodiment in FIG. 5, the PTAC sleeve or receptacle **40** may be hard-wired.

[0021] The wirelessly powered PTAC may be stowed or assembled with the PTAC receptacle in use and separated or deployed away from the PTAC receptacle when replacing/repairing/installing

the PTAC **60** and/or when installing the PTAC receptacle **40** within the wall **11**. When the PTAC is deployed away from the PTAC receptacle **40** as shown in the one embodiment in FIG. **3** (e.g. deployed position), the wireless power transmitter **44** may be exposed. When it is desirable to stow or install/assemble the PTAC **60** with the PTAC receptacle **40** as shown in FIGS. **1** and **5** (e.g. stowed or in use position), the wirelessly powered PTAC **60** and the wireless power receiver **62** and/or PTAC **60** may be placed with the PTAC receptacle **40** in alignment with wireless power transmitter pad **44**, and power may be supplied to the wirelessly powered PTAC. It will be appreciated that PTAC **60** and PTAC receptacle **40** may each be specifically designed such that when PTAC **60** is received within PTAC receptacle **40**, wireless power transmitter **44** and wireless power receiver **62** are aligned with one another to maximize power transfer efficiency.

[0022] Now turning to FIG. **2**, PTAC wireless power system **12** may be under the control of a controller **50** that receives inputs from a number of components and drives a number of components in response thereto. Controller **50** may, for example, include one or more processors **52** and a memory **54** within which may be stored program code for execution by the one or more processors. The memory may be embedded in controller **50**, but may also be considered to include volatile and/or non-volatile memories, cache memories, flash memories, programmable read-only memories, read-only memories, etc., as well as memory storage physically located elsewhere from controller **50**, e.g., in a mass storage device or on a remote computer interfaced with controller **50**. Controller **50** may also be implemented at least in part using discrete circuit logic, as will be appreciated by those of ordinary skill having the benefit of the instant disclosure.

[0023] As shown in FIG. **2**, controller **50** may be powered by a power supply **56**, e.g., an AC-DC power supply that is coupled to line power **58** (e.g., 120-240 VAC, as may be provided by a US residential/commercial electrical circuit, 15-30 Amps, 265-277 VAC as may be provided by some US commercial electrical circuits, or other currents/voltages as may be supplied in other countries), via an electrical outlet and plug, or alternatively, a hard-wired connection. Line power **58** also supplies wireless power transmitter **44** with power. Controller **50** may control wireless power transmitter **44** to selectively activate/deactivate the wireless power transmitter, to regulate the power output of the wireless power transmitter, to communicate data to and/or receive data from wirelessly powered PTAC **60**, to determine the power output required by the wirelessly powered PTAC, etc. In the illustrated embodiment, wireless power transmitter **44** is compatible with the Ki Kitchen Cordless Standard developed by the Wireless Power Consortium, although other wireless power or charging standards may be used in other embodiments. It will be appreciated that the control over wireless power transmitter **44** by controller **50** to emit a wireless power signal would be well within the abilities of those of ordinary skill having the benefit of the instant disclosure.

[0024] A wirelessly powered PTAC **60** may include a wireless power receiver **62** that, when positioned proximate wireless power transmitter **44**, receives a wireless power signal to supply power to the wirelessly powered PTAC. Wirelessly powered PTAC **60** may also include a controller **64** to operate wireless power receiver **62**, as well as to perform other appliance-related functions. Power received by wireless power receiver **62** may be used to also power one or more electrical loads **66**, e.g., one or more motors, compressors, electric heating elements, fans, heat pumps, condensers, evaporators, expansion valves, displays, controls, switches, dampers, slinger wheels, etc., as well as controller **64** itself.

[0025] In some embodiments, wirelessly powered PTAC **60** may include a user interface **68** to operate the appliance, and, beyond the supply of wireless power, may operate completely independently from PTAC wireless power system **12**. In other embodiments, however, PTAC wireless power system **12** may be functionally integrated with wirelessly powered PTAC **60**, e.g., such that a user interface **70** of PTAC wireless power system **12** is used to display information received from wirelessly powered PTAC **60** and/or control one or more functional operations of wirelessly powered PTAC **60** beyond simply controlling the supply of power to the wirelessly powered PTAC, such that a user interface **68** of wirelessly powered PTAC **60** is used to display

information received from the PTAC wireless power system **12** and/or control one or more functional operations of PTAC wireless power system **12**. In some embodiments, for example, wirelessly powered PTAC **60** may omit any separate user interface, and may be operated through user interface **70** of PTAC wireless power system **12**. In some embodiments, for example, PTAC wireless power system **12** may omit the user interface, and may be operated through user interface **68** of wirelessly powered PTAC **60**. Communication between PTAC wireless power system **12** and wirelessly powered PTAC **60** may be over a Near Field Communication (NFC) wireless link as supported by the Ki standard, or via a separate wired or wireless network.

[0026] It may also be desirable to provide one or more sensors **72** for sensing various states associated with the PTAC wireless power system **12**. In some embodiments, for example, a temperature sensor and/or current sensor may be used to monitor wireless power transfer and/or prevent overheating. Other suitable sensors will be appreciated by those of ordinary skill having the benefit of the instant disclosure.

[0027] It should be appreciated that wireless power is distinguished from wireless charging in the context of the present disclosure, as while a PTAC wireless power system may in some instances provide power to charge a battery of a PTAC wirelessly coupled to the wireless power system, such a use is secondary to providing wireless power to operate the PTAC, i.e., to provide the primary power supply to the PTAC during its active use. As such, a wirelessly powered PTAC in many instances may lack or not include any battery or other power storage element capable of independently providing sufficient power to operate the wirelessly powered PTAC, such that the wirelessly powered PTAC is effectively inoperable unless coupled to a wireless or wired power source. For many wirelessly powered PTACs, high power draw electrical loads such as electric motors and/or compressors may be used, and to the extent any power storage element is present on such a wirelessly powered PTAC, it is incapable of sufficiently powering such high power draw electrical loads, such that the wirelessly powered PTAC is principally powered through a PTAC wireless power system as described herein.

[0028] A PTAC wireless power system consistent with the invention may therefore generally be considered to include at least a sleeve and/or receptacle **40** that is mountable to a wall **11**, or portion thereof, along with a wireless power transmitter **44** supported by the receptacle **40** and configured to generate a wireless power signal to power a wirelessly powered appliance **60** (e.g. PTAC) positioned proximate thereto. A receptacle, or portions thereof, in particular, may be considered to be mountable to a wall when the receptacle is sized and configured, and in some instances, includes suitable mounting hardware, for securing (e.g. through, within) the receptacle to the wall. For example, a mounting system may be used in some embodiments to secure/support/couple the PTAC receptacle to the wall. In some embodiments, the mounting system (e.g. bracket, extension, grill) may include one or more portions of the wireless transmitter.

[0029] Mounting may occur through the use of fasteners and/or adhesives, and mounting may be made to any structure of a wall and/or structure providing suitable support (e.g. mounting system). Mounting, for example, may be made through the use of threaded fasteners that extend into the surface of a wall in some embodiments. In addition, in some embodiments, additional mounting hardware may be used to mount a receptacle to a wall, e.g., one or more mounting brackets, and in some embodiments a receptacle may be removably attachable to a mounting bracket such that a bracket may first be mounted to a wall and the receptacle thereafter attached to the mounting bracket to complete the installation. The receptacle and/or mounting bracket may also include, in some embodiments, a structure suitable for hiding a portion of a power cord, e.g., when the receptacle is secured to a wall directly adjacent an electrical outlet such that any extra length of power cord can be hidden from view. In some embodiments, the receptacle may be secured directly to a portion of the wall outlet/box.

[0030] The PTAC receptacle may be a variety of shapes, sizes, quantities, and constructions and still be within the scope of the invention. In some implementations, the PTAC sleeve/receptacle **40**

may include one or more members **42**. The one or more members **42** may include the wireless transmitter **44**, or portions thereof. The one or more members may position the wireless transmitter **44** to communicate and/or align with the wireless power receiver **62** when the PTAC is stowed. The one or more members **42** may engage/couple/receive/support at least a portion the wirelessly powered PTAC **60**. The one or more members may engage a portion of the wall **11** and/or define at least a portion of a through opening **45** within the wall. The one or more members **42** may define at least a portion of an inner periphery **43** defining the through opening **45** therein. As shown in the one embodiment, the one or more members **42** may include a top wall **42a**, bottom wall **42b**, and opposing side walls **42c** interconnecting the top wall and the bottom wall. The through opening **45** of the receptacle **40** may receive the wirelessly powered PTAC **60**, or portions thereof, when in the use/stowed configuration. Although the receptacle defines a 360 degree through opening, the one or more members may define a portion of the receptacle less than 360 degrees in some embodiments. The PTAC receptacle **40** and/or member(s) may be of a variety of shapes, sizes, quantities, and/or constructions and still be within the scope of the invention. The PTAC receptacle may be capable of receiving one or more PTACs of a variety of characteristics. For example when replacing or installing, the PTAC receptacle may stow/receive/engage a variety of different/replacement PTACs having different sizes, shapes, electrical loads, user interfaces, controls, constructions, etc. Alternatively when replacing the PTAC, the replacement PTAC may have one or more characteristics the same (e.g. shapes, sizes, electrical loads).

[0031] In some implementations, the PTAC receptacle **40**, system **12**, and/or member(s) **42** may include at least one wireless transmitter **44**. As shown in the one embodiment in FIGS. **1**, **3**, and **4**, the bottom wall **42b** or member **42** may include the wireless transmitter **44**. Correspondingly, the bottom, or adjacent thereto, of the wirelessly powered PTAC may include the wireless power receiver **62**. As shown when in the stowed position or aligned, the wireless power transmitter may be upwardly-facing to oppose a downwardly-facing wireless power receiver of the wirelessly powered PTAC. As shown in the one embodiment in FIG. **5**, the side wall **42c** or member **42** may include the wireless transmitter **44**. Correspondingly, the side, or adjacent wall thereto, of the wirelessly powered PTAC **60** may include the wireless power receiver **62**. As shown when in the stowed position or aligned, the wireless power transmitter may be side-facing to oppose a side-facing wireless power receiver of the wirelessly powered PTAC. Other positions of the wireless transmitter **44** and/or wireless power receiver **62** are contemplated. For example, the top wall of the receptacle **40** may include the wireless transmitter **44** and the PTAC top wall may include the wireless power receiver **62**. Whereby when in the stowed position or aligned, the wireless power transmitter may be downwardly-facing to oppose an upwardly-facing wireless power receiver of the wirelessly powered PTAC. Further for example, a mounting bracket or accessory for the PTAC receptacle may include the wireless transmitter.

[0032] In some implementations, the wireless power transmitter may be inwardly-facing and/or direct a wireless power signal inwardly to supply power to the wirelessly powered PTAC. The wireless power transmitter may face inwardly or radially inward from the inner periphery **43**, member **42**, and/or receptacle **40** as shown in the embodiments. The wireless power transmitter **44** may face in a direction substantially perpendicular to the telescoping/deploying/stowing direction of the PTAC **60** relative to the PTAC wall receptacle **40** and/or longitudinal axis of the through opening **45**. The wireless power transmitter may face longitudinally or parallel to the longitudinal axis of the through opening **45** defined by the one or more members **45** and/or inner periphery **43**. Although not shown, wireless power transmitter **44** may face in a direction substantially parallel to the telescoping/deploying/stowing direction of the PTAC **60** relative to the PTAC wall receptacle **40**. For example, when in the stowed position or aligned, the wireless power transmitter may be front-facing (e.g. away from the outside/grill, towards front end of PTAC receptacle, towards receiving end or opening of the through opening, towards deploying direction) to oppose a rear-facing (e.g. towards the rear end of the PTAC receptacle, towards grill, towards rear end of PTAC,

in stowing direction) wireless power receiver of the wirelessly powered PTAC.

[0033] In some implementations, the PTAC wireless power system **12**, PTAC receptacle **40**, member(s) **42**, PTAC **60**, and/or wireless transmitter **44** may be powered in a variety of ways or constructions. In the one embodiment shown in FIGS. **1**, **3**, and **4**, the wireless transmitter **44** and/or system **12** is supplied with power by a power cord **48** that is plugged into an electrical outlet **38** housed in a junction box **114** positioned on a wall **11**. In the one embodiment shown in FIG. **5** illustrates another variation, whereby rather than being powered by a power cord plugged in to electrical outlet **38**, PTAC wireless power system **12** may be hard-wired to an electrical circuit (e.g. junction box), as represented by wire **148** shown running within wall **11**.

[0034] In use, PTAC **60** is stowed with the PTAC wireless power system **12** or PTAC receptacle **40** in the configuration illustrated in FIGS. **1**, **4**, and **5**, with PTAC **60** nested/coupled/stowed (e.g. releasably) with PTAC receptacle **40**, or portions thereof. When it is desired to power a wirelessly powered PTAC **60**, when in the use or stowed position/configuration (FIGS. **1**, **4**, and **5**) the wirelessly powered PTAC **60** with its wireless power receiver **62** is aligned with wireless power transmitter **44** of the PTAC receptacle **40**. Power may then be supplied to the wirelessly powered PTAC **60**, e.g., automatically after detection of wireless power receiver **62** by wireless power transmitter **44**, or manually after user selection of an “on” function supported by PTAC wireless power system **12** or wirelessly powered PTAC **60**. If the wirelessly powered PTAC **60** is desired to be removed from the stowed position to the deployed position as illustrated in FIG. **3**, wirelessly powered PTAC **60** may be removed from PTAC receptacle **40** (e.g. misalignment of the receiver and transmitter).

[0035] In some implementations, the wireless transmitter and receiver may communicate with each other to negotiate appropriate power transfer. Whereby a PTAC receptacle may be used with different types of PTACs (e.g. having different power requirements), and automatically adjust to supply the appropriate power for the corresponding PTAC installed therein. For example in a hotel application that may have 265/277 VAC power, a PTAC receptacle **40** in the room may be installed with a PTAC **60** for use with 208/230 VAC and have the wireless power system **12** adjust the power supplied to meet the requirements of these (previously-incompatible) PTACs.

[0036] It will be appreciated that, while certain features may be discussed herein in connection with certain embodiments and/or in connection with certain figures, unless expressly stated to the contrary, such features generally may be incorporated into any of the embodiments discussed and illustrated herein. Moreover, features that are disclosed as being combined in some embodiments may generally be implemented separately in other embodiments, and features that are disclosed as being implemented separately in some embodiments may be combined in other embodiments, so the fact that a particular feature is discussed in the context of one embodiment but not another should not be construed as an admission that those two embodiments are mutually exclusive of one another. Various additional modifications may be made to the illustrated embodiments consistent with the invention. Therefore, the invention lies in the claims hereinafter appended.

Claims

1. A PTAC wireless power system comprising: a PTAC receptacle configured to mount to an outer wall; and a wireless power transmitter supported by the PTAC receptacle and configured to generate a wireless power signal to power a wirelessly powered PTAC positioned proximate thereto.
2. The PTAC wireless power system of claim 1 wherein the wirelessly powered PTAC is movable between a deployed position and a stowed position relative to the PTAC receptacle.
3. The PTAC wireless power system of claim 2 wherein the wireless power transmitter is upwardly-facing to oppose a downwardly-facing wireless power receiver of the wirelessly powered PTAC when the wirelessly powered PTAC is in the stowed position.

- 4.** The PTAC wireless power system of claim 2 wherein the wireless power transmitter is downwardly-facing to oppose an upwardly-facing wireless power receiver of the wirelessly powered PTAC when the wirelessly powered PTAC is in the stowed position.
- 5.** The PTAC wireless power system of claim 2 wherein the wireless power transmitter is side-facing to oppose a side-facing wireless power receiver of the wirelessly powered PTAC when the wirelessly powered PTAC is in the stowed position.
- 6.** The PTAC wireless power system of claim 1 further comprising a power cord configured to supply power to the wireless power transmitter from a wall-mounted electrical outlet.
- 7.** The PTAC wireless power system of claim 1 wherein the PTAC receptacle is configured to supply power to the wireless power transmitter from a hard-wired connection of an electrical circuit.
- 8.** The PTAC wireless power system of claim 1 further comprising a user interface and a controller configured to operate the user interface and the wireless power transmitter, wherein the controller is further configured to control a functional operation of the wirelessly powered PTAC using the user interface.
- 9.** The PTAC wireless power system of claim 1 wherein the wirelessly powered PTAC includes a wireless power receiver, a user interface, and a controller configured to operate the user interface and the wireless power receiver, wherein the controller is further configured to control a functional operation of the PTAC receptacle using the user interface.
- 10.** The PTAC wireless power system of claim 1 wherein the wireless power transmitter is configured to supply power to a wirelessly powered PTAC including an electric motor and/or a compressor and lacking a power storage element sufficient to otherwise power the wirelessly powered PTAC.
- 11.** The PTAC wireless power system of claim 1 wherein the PTAC receptacle includes one or more members defining at least a portion of a through opening, wherein the wireless power transmitter is supported by the one or more members of the PTAC receptacle.
- 12.** The PTAC wireless power system of claim 11 wherein the one or more members includes a top wall, a bottom wall, and opposing side walls interconnecting the top wall and the bottom wall.
- 13.** The PTAC wireless power system of claim 1 further comprising a controller configured to regulate a power output of the wireless power transmitter.
- 14.** The PTAC wireless power system of claim 13 wherein the controller is configured to determine the power output required by the wirelessly powered PTAC and control the wireless power transmitter to output a wireless power signal that meets the power output required by the wirelessly powered PTAC.
- 15.** The PTAC wireless power system of claim 1 wherein the wireless power transmitter is inwardly-facing to oppose an outwardly-facing wireless power receiver of the wirelessly powered PTAC.
- 16.** A PTAC wireless power system comprising: a wirelessly powered PTAC having a wireless power receiver; and a PTAC receptacle having a wireless power transmitter configured to generate a wireless power signal in communication with the wireless power receiver of the wirelessly powered PTAC to power the wirelessly powered PTAC positioned proximate thereto.
- 17.** The PTAC wireless power system of claim 16 further comprising a user interface and a controller configured to operate the user interface and the wireless power transmitter, wherein the controller is further configured to control a functional operation of the wirelessly powered PTAC using the user interface.
- 18.** The PTAC wireless power system of claim 16 wherein the wirelessly powered PTAC includes a wireless power receiver, a user interface, and a controller configured to operate the user interface and the wireless power receiver, wherein the controller is further configured to control a functional operation of the PTAC receptacle using the user interface.
- 19.** The PTAC wireless power system of claim 16 further comprising a controller configured to

regulate a power output of the wireless power transmitter.

20. The PTAC wireless power system of claim 19 wherein the controller is configured to determine the power output required by the wirelessly powered PTAC and control the wireless power transmitter to output a wireless power signal that meets the power output required by the wirelessly powered PTAC.

21. A wirelessly powered PTAC comprising: a PTAC unit configured to couple with a PTAC receptacle between a stowed position and a deployed position; and a wireless power receiver supported by the wirelessly powered PTAC and configured to receive a wireless power signal provided by a wireless power transmitter of the PTAC receptacle to power the wirelessly powered PTAC when positioned in the stowed position.
